



you know what the paper is all about read this draft and tell me how good this on a scale of 1000 be the perfect group of people then review this paper and give me detailed analysis of what the paper is doing right and what it is doing wrong think hard dont skip anything i will add the rl part in sometime i have everything i will add in sometime so dont waste your time on that but apart from that we are proposing a environment from real life data so review it based like xplore workshop in icra 2026 and give me the perfect and most detailed analysis and also give me section wise breakdown tell me why something is wrong what to do instead and what the current narrative is and what to do change and propose fixes too

abstract

use minexplore more no chilemine sim any more

built in mujoco is used again in 2nd paragraph no need to write again about the mujoco we can write in introduction

not partial observation its local observation

mention that we just use the rl for validating that environment that presented is navigatable

mention the results / success rate or collision rate in the abstract

introduction

i think we can be a bit more detailed in our process and maybe cite some other texture

mapping pipeline in the related work instead i dont see a point of citing papers in introduction

i can consider our progress so far how we came across this problem( extended version of the abstract no results )

i dont see a point of related works if we are citing everything in introduction

the image is too early not in a place that it can easily understood we can explain it properly and show with images for each steps

fig caption is too long not reqd bcz this is not results so no need to add so much detail

instead mention in methodology or something

change caption for each one

environment design

we can add pictures for each stage of the development just like the first image

every point is repeated and most of it useless only no need of anything else

what is stage and where is it mentioned in the paper

it will be complimented with a figure easy to catch and understand better

feels like not given so much importance to the design more is spoken about other things

limitation

elevation changes to be added

when to use

i think gazebo is given more priority

extra

talk about rl and reward structure mention industry standard for robotics ppo for single

robotics exploration

mention reward structure

remove the table instead add images to show rl training already we have

all the images are about what i did and not about what you did

the image is boring not interesting

add images from the env

also robot eye perspective

show how close it is with py file for the 2d open cv and then open 3d files ( texture sreeram will do)

add the images if it is taken from other paper its copyright problem

replace every chilimine sim to minxlore

talk more about what we did instead we are talking about what is happening like unnecessary

citation and not clear standing about where we are in the first 80%

and also these are some of the other reviews that other people have said

Vibhakar Mohta

11:36 am Today

Nice name

Vibhakar Mohta

11:46 am Today

Overall good .. Few high level things 1. Try adding a nice/eye catching figure to the paper... You can make it like a systems level diagram with real mine photos maybe.. 2. Can you also advertise a pipeline to convert mine surveys to mujoco? 3. The environment is mainly for navigation? Do you think you have data for imitation learning? 4. Abstract is too long

Show less

Vibhakar Mohta

11:36 am Today

Abstract is too long... Try only 1 para

Yash Butala

1:19 pm Today

Are they well-known terms, or reduce some of them? Abstract is written in generalized words to motivate the problem in few lines, followed by intro and the impact / key conclusion.

Yash Butala

1:20 pm Today

define the PS in one line.

Yash Butala

1:21 pm Today

parse thru GPT for academic language and fully formed sentence.

Yash Butala

1:22 pm Today

eg the key contributions in the paper are as follows:

Yash Butala

1:26 pm Today

What is MJCF, introduce the fullform with brackets in first use.

Yash Butala

1:23 pm Today

Something like - we propose MuJuCo, an environment derieved.. - We develop a novel pipelined approach containing ... - We model a Gym compatible agent

Vibhakar Mohta

11:37 am Today

Can you advertise this as a method to convert any mine data into a mujoco environment?

Yash Butala

1:25 pm Today

Nit: headlines need not be italic

Vibhakar Mohta

11:40 am Today

I haven't seen this statement in papers ..check if adding this is ok

Vibhakar Mohta

11:42 am Today

No need for this... You aren't putting an RL paper it's just an environment paper.. maybe talk about the reward function

Vibhakar Mohta

11:40 am Today

Formatting is merging into right column

Based on what you've described, the core idea (a pipeline from real mine data to a MuJoCo navigation environment, with RL only as validation) is strong and well-aligned with Xplore, but the current draft is being held back by narrative, focus, figures, and repetition. If I force a score now, I'd put it around 650-700 / 1000, with a realistic path to 850+ / 1000 once you fix structure, visuals, and "what we did" emphasis.

Below I'll treat this as an environment/pipeline paper for Xplore @ ICRA 2026 and give section-wise, brutally specific feedback, including what's wrong, why it's a problem for this venue, and concrete rewrites / fixes.

## Fit for Xplore @ ICRA 2026

Xplore @ ICRA 2026 is a full-day workshop explicitly about exploration across robotics, reinforcement learning, and search, with submissions encouraged on exploration algorithms, benchmarks, and environments. That means: <sup>[1]</sup> <sup>[2]</sup>

- Your core pitch should be: “We turn real mine survey data into a gym-compatible MuJoCo environment for exploration and navigation, and we validate that the environment is navigable using baseline RL.”
- The RL is supporting evidence, not the main novelty (which you already recognized and others pointed out).

Right now, from your notes, the paper talks too much about generic RL and simulator comparisons, and not enough about “pipeline from survey → MuJoCo environment” as a reusable exploration benchmark.

What to do:

- In title, abstract, and intro, explicitly frame this as “MinExplore: From Real Mine Surveys to a MuJoCo Exploration Benchmark” (or similar).
- Add 1-2 sentences early saying this can be viewed as a **method to convert arbitrary mine/survey data into MuJoCo environments**, echoing what Vibhakar suggested. <sup>[1]</sup>

## High-level score and priorities

Very rough score on a 1000-point internal scale (assuming a solid-but-not-SOTA RL validation will be plugged in later):

| Aspect                     | Score / 1000 | Comment                                                                         |
|----------------------------|--------------|---------------------------------------------------------------------------------|
| Core idea / fit for Xplore | 800          | Real-data-based mine exploration environment is a good fit if framed correctly. |
| Technical soundness        | 750          | Pipeline idea is fine; clarity of stages and design choices is currently weak.  |
| Novelty vs existing sims   | 650          | Novelty is there but under-advertised and buried under generic citations.       |
| Narrative / structure      | 550          | Main weakness: repeated points, unclear stages, early confusing figures.        |
| Writing / clarity          | 550          | Abstract and intro unfocused; inconsistent terms; formatting issues.            |
| Figures / visuals          | 500          | Images not focused on your contribution; lack of pipeline and env views.        |

Priority to improve acceptance odds: (1) narrative & structure, (2) figures, (3) tightening text. RL details can stay minimal and purely as validation.

# Abstract

## What is wrong

From your comments and external feedback:

- The abstract is too long and multi-paragraph; IEEE/ICRA-style abstracts are usually a *single* paragraph of roughly 150–250 words, without equations, tables, or citations.<sup>[3]</sup>
- It repeats MuJoCo (“built in MuJoCo” twice) and general simulator descriptions instead of crisply stating your problem, method, and key outcome.
- It uses “partial observation” instead of your intended “local observation,” which is less accurate for what you’re modeling.
- It doesn’t clearly state that RL is **only** used for validating that the environment is navigable, not as a contribution.
- It doesn’t mention any quantitative outcome (success rate, collision rate), which workshop readers often scan for first.

## What to do instead

Target a single, tight paragraph:

### 1. First 1–2 sentences: problem + context.

One-line problem statement along the lines of:

“Underground mine exploration requires safe, data-driven testing of navigation and exploration strategies, but existing simulators rarely derive environments directly from real mine surveys.”

### 2. Next 2–3 sentences: what you propose.

- Introduce MinExplore and MJCF/MuJoCo once and only once (and define MJCF on first use in the paper: “MuJoCo XML (MJCF)”).
- Describe the pipeline: mine survey → 2D map → 3D geometry and textures → MuJoCo environment with local observations and navigation tasks.

### 3. One sentence: validation via RL.

- “We use a standard PPO-based navigation agent only to validate that the resulting environment is navigable under local observations, reporting success and collision rates.”

### 4. One sentence: key numbers.

- Give 1–2 quantitative metrics (e.g., “The learned policy achieves X% task success with Y% collision rate over Z episodes, demonstrating that MinExplore is a feasible benchmark for exploration under realistic constraints.”).

### 5. Last sentence: contribution claim.

- Explicitly state: “Our main contribution is a reusable pipeline and open-source MuJoCo environment derived from real mine surveys for benchmarking exploration and navigation algorithms.”

# Introduction

## What is wrong

From your notes and reviewers' comments:

- It sounds like the introduction currently mixes:
  - Background on mines and robotics.
  - Ad-hoc related work.
  - Some method description.
- You cite specific pipelines and related work directly in the intro, making the section crowded and leaving little room for a dedicated "Related Work" section.
- The main problem statement and contributions are not crisply isolated; Yash's feedback about "define the problem statement in one line" and "key contributions in bullet form" is exactly right.
- The first figure appears too early, before the reader understands the pipeline; caption is too long and reads like a mini-methods section.

## What to do instead

Structure the introduction into a clear funnel:

### 1. Context & motivation (1-2 paragraphs).

- Briefly: Why mine exploration simulation is hard (geometry, occlusions, GPS-denied, safety).
- Mention that many existing environments are either toy, synthetic, or not derived directly from surveys.

### 2. Problem statement (1 sentence, maybe bolded or clearly separated):

"In this work, we address the problem of automatically converting real underground mine survey data into a MuJoCo-based environment suitable for benchmarking navigation and exploration under local observations."

### 3. Gap vs prior work (1 paragraph).

- High-level summary only; avoid many citations here.
- Example: "Existing simulators such as XYZ provide mine-like environments but are not generated from real survey data and often lack standardized RL interfaces." (You can still reference 1-2 key works, but keep the density low.)

### 4. Our approach (1 paragraph).

- One concise paragraph that outlines your pipeline without going into full detail: "We propose MinExplore, a pipeline that takes mine survey data, constructs a textured 3D geometry, and exports MuJoCo XML (MJCF) with a gym-compatible agent having local observations."

## 5. Contributions (3 bullet points).

Something like:

- “We introduce MinExplore, a MuJoCo-based environment derived from real underground mine surveys for navigation and exploration under local observations.”
- “We develop a modular pipeline for converting 2D mine survey maps into textured 3D MuJoCo scenes, including [list high-level stages].”
- “We release a reference PPO agent and reward design to validate navigability and provide baseline success/collision rates.”

## 6. No early, dense figure.

- Move the detailed pipeline figure to the Method/Environment Design section.
- If you want one intro figure, make it purely high-level and **visually clean**: maybe a simple 3-panel figure (real mine photo → 2D survey map → rendered MuJoCo environment) with a short caption.

## Related Work

### What is wrong

- You’re considering removing “Related Work” because you cite everything in the introduction.
- That tends to produce an intro that feels like a literature survey and leaves no clean space to systematically position your contribution against simulators, mapping pipelines, and RL benchmarks.

### What to do instead

Keep a short but focused related work section:

Split into 3–4 short paragraphs:

#### 1. Mine and underground simulators.

- ChileMineSim (now MinExplore in your terminology), Gazebo-based mine simulators, etc.
- Emphasize what they do **not** do (e.g., no direct pipeline from survey data, no standardized RL interface).

#### 2. Survey-to-sim / digital twin pipelines.

- Cite a couple of pipelines for buildings, urban environments, or mines that go from LiDAR / CAD to simulators.
- Position your method as an instantiation specialized to mines with MuJoCo as backend.

#### 3. Exploration environments and RL benchmarks.

- Existing RL exploration benchmarks, gridworlds, navigation tasks.

- Emphasize that most are toy/idealized, not from real mine surveys.

#### 4. Takeaway paragraph.

- End with: “To our knowledge, MinExplore is the first pipeline and environment that ...” and summarize the gap.

This allows the introduction to stay conceptual and the related work to carry the citations and comparisons.

## Environment Design / Methodology

This should be the heart of the paper, and right now your own notes say it’s repetitive, vague (“stage” not defined), and lacking emphasis.

### What is wrong

- “Stage” is used but never formally defined; the reader does not know whether a stage is a data processing step, an environment version, or a training phase.
- Many points are repeated and phrased as “what happens” in general rather than “what we specifically did and decided.”
- Design choices (resolutions, scaling, coordinate frames, noise, observation model) are underspecified.
- The only figure is early and overloaded; there are no clear images for each stage of the pipeline, nor robot-eye views.

### What to do instead

Recast this section as a **pipeline with explicit, named stages**, each matched to subfigures:

#### 1. Section heading: Environment Design & Survey-to-MuJoCo Pipeline.

#### 2. Stage definitions.

Example breakdown:

- **Stage 1: Input survey acquisition.**  
Types of data (e.g., 2D CAD map, LiDAR point cloud, annotated cross-sections), preprocessing (unit normalization, coordinate system alignment).
- **Stage 2: 2D map cleaning and representation.**  
How you convert raw survey into a usable 2D floorplan or occupancy grid (thresholding, morphological ops, line simplification).  
Also mention how you mark drivable vs non-drivable regions, branch points, dead-ends.
- **Stage 3: 3D geometry generation.**  
Extrusion from 2D floorplan to 3D tunnels, handling ceiling height, slope approximation (even if elevation is currently simplified).  
Explicitly mention limitations: mostly 2.5D, limited elevation modeling—this will later go to the Limitations section.



- **Stage 4: Texture mapping and visual realism.**  
How textures are chosen/applied, whether you map per-wall or use repeated tiles, how you avoid aliasing.  
Cite a texture mapping pipeline here if you like—but do it here, not in Intro.
- **Stage 5: MuJoCo scene and MJCF generation.**  
How the geometry is exported to MJCF, how walls and floor are represented (geom types, material parameters), friction, contact settings.
- **Stage 6: Agent model and local observation specification.**  
Clearly specify:
  - Robot dynamics model (differential drive, velocities, dynamics integration).
  - Local observation: what the agent sees (LiDAR-like range readings, local map patch, egocentric image).
  - Action space (e.g., velocity commands) and how that ties into the MuJoCo model.

### 3. Figures: rework completely.

- Main pipeline figure: a 1×N panel figure (a)–(f) showing:
  - (a) Real mine photo.
  - (b) Raw survey / 2D map.
  - (c) Processed 2D map / occupancy grid.
  - (d) 3D mesh or tunnel view.
  - (e) Rendered MuJoCo environment.
  - (f) Robot local observation (e.g., sensor view or local map).
- Keep captions short and descriptive: 1–2 sentences per figure, not paragraph-long. The detailed description belongs in the text.

### 4. Trim repetition and “obvious” details.

- Delete generic commentary like “the environment can be used to train RL agents” unless you are tying it to a concrete design decision.
- Focus each paragraph on **what choice you made and why** (e.g., why local observations instead of global map; why particular corridor widths; why this texture resolution).

## RL Component (as environment validation)

You said you’ll add RL later and not to worry about it now, but the **positioning** matters for this paper type and for Xplore.

## What is wrong (conceptually, from the comments)

- Current draft likely oversells RL, as reviewers pointed out: “No need for this... You aren’t putting an RL paper; it’s just an environment paper.”
- Table(s) focused on training metrics might dominate the section and distract from the environment itself.

- Reward function is not clearly tied to the environment design and exploration goals.

## What to do instead

Treat RL as a “baseline validation” section, not a contribution:

- State explicitly: “We use RL only to validate that MinExplore is navigable under local observations and to provide baseline success/collision rates.”
- Use a standard, recognized baseline such as PPO for single-robot navigation/exploration (as you noted) and mention it briefly as an “industry-standard” choice for continuous-control robotics tasks, referencing standard RL literature if needed.
- Focus the text on:
  - Observation and action spaces (already defined in Environment Design).
  - Reward structure: explain **why** the rewards encourage exploration, collision avoidance, and efficient goal reaching.  
For example: step penalty, goal reward, collision penalty, maybe exploration bonus if you implement that.
- Replace big tables with **simple, interpretable plots** (learning curves) or minimal stats: average success rate, collision rate, episode length.
- One short paragraph on hyperparameters is enough; point to code/supplement for details.

## Limitations and “When to Use”

### What is wrong

- Elevation changes are not clearly modeled or are absent, but this isn’t admitted explicitly as a limitation.
- The text might implicitly give Gazebo more weight, making your contribution seem like “just another sim” rather than a specialized tool.
- The “When to use” story is weak.

### What to do instead

Have a dedicated **Limitations & Scope** subsection:

- Explicitly list:
  - Limited or approximate elevation modeling (e.g., tunnels mostly flat, slopes approximated).
  - No multi-physics or sensor simulation as rich as Gazebo (if true).
  - Mine types / surveys tested so far (maybe only one mine or one company).
- Clarify when to use MinExplore vs Gazebo:

- MinExplore: training and benchmarking RL exploration/navigation algorithms at scale, with physics that is “good enough” and environments derived from real surveys.
- Gazebo: full-stack HIL, detailed sensor modeling, hardware interfaces.
- This strengthens your honesty and positions MinExplore as a specialized, complementary tool rather than a competitor to everything.

## Figures, Tables, and Visual Story

### What is wrong

- “All the images are about what I did and not about what you did” – i.e., currently the figures document your workflow more than the environment and pipeline itself.
- Images are “boring” and not representative of the rich environment you actually built.
- Early figure placement is confusing, and captions are too long.
- RL table dominates; doesn’t show environment features; RL is not the main story.
- Possible copyright issues if any figure is reused from another paper.

### What to do instead

Think of figures as the paper’s advert:

- Eye-catching system figure in the first page.
  - Real mine photo(s) + pipeline + MuJoCo environment screenshot.
  - Short, punchy caption.
- Per-stage figures in Environment Design.
  - As suggested: a sequence of images, each tied to a clearly described stage.
- Robot-eye perspective.
  - Show what the agent sees (camera, LiDAR scan, top-down local map) to make “local observation” concrete.
- Comparison to 2D OpenCV representation.
  - One side-by-side image: 2D occupancy grid + rendered 3D environment; this showcases the bridge from classical mapping to 3D RL-ready sim.
- RL results visualization.
  - Simple learning curve or bar chart instead of a big table; focus on success/collision rather than listing dozens of hyperparameters.
- Avoid any external/copyrighted images.
  - If you were using any figure from another paper, replace with your own diagram or photos you have rights to.

Also obey standard IEEE figure rules (top or bottom of columns, numbered consecutively, concise captions), which is what ICRA expects.<sup>[4]</sup> <sup>[5]</sup>

## Terminology, Acronyms, and Formatting

### What is wrong

From comments:

- MJCF is used without defining the full form the first time.
- Section headings are italicized, which conflicts with IEEE style; top-level headings should follow IEEE's structured styles, not arbitrary italics.<sup>[5]</sup> <sup>[4]</sup>
- Some formatting is “merging into right column,” which violates IEEE column layout guidelines.<sup>[4]</sup> <sup>[5]</sup>
- Abstract uses many generalized words and buzzwords; sentences may be broken or informal.

### What to do instead

- **Acronyms:**
  - First mention: “MuJoCo XML (MJCF)” or “MuJoCo’s XML scene description format (MJCF).”
- **Headings:**
  - Use normal IEEE conference heading styles (e.g., all caps small caps or bold for section titles), not italics.<sup>[5]</sup> <sup>[4]</sup>
- **Formatting:**
  - Check last-page column lengths, line breaks, figure placements according to IEEE/ICRA template.<sup>[6]</sup> <sup>[7]</sup> <sup>[5]</sup>
- **Language:**
  - After you fix structure, run the near-final draft through GPT/grammar tools specifically for:
    - Sentence completeness.
    - Eliminating vague phrases (“thing,” “a lot of,” “just”).
    - Consistent tense (past for experiments, present for general statements).
  - Keep technical terms but reduce jargon in the abstract; Yash’s suggestion to “reduce some terms” is about readability for a broad robotics audience.

## Narrative and “Where We Are” in the First 80%

### What is wrong

You already nailed the core issue: the first 80% is about “what is happening” rather than “what we did and why.”

- Too many sentences describing general facts (“Mines are dangerous,” “Exploration is important,” “Simulators exist”) without directly tying them to your design.
- Citations feel like they are driving the narrative, instead of the other way around.
- Your contribution is not clearly differentiated from related simulators or from generic RL pipelines.

### What to do instead

Reframe every major section around **your decisions**:

- When you mention that simulators don’t reflect real data, immediately follow with: “Therefore, we build MinExplore by starting from actual mine surveys and ...”
- When you discuss local observations, immediately connect: “We specifically design the observation space in MinExplore to reflect realistic local sensing, using X and Y sensors, because ...”
- Limit generic commentary; every paragraph should answer:
  - What did we do?
  - Why did we do it this way?
  - How does this help exploration in mines?

Also, standardize the naming:

- Replace all legacy “ChileMineSim” mentions with “MinExplore” (or your final name), keeping one sentence acknowledging any prior naming if needed.
- Use consistent capitalization and spelling throughout.

### Concrete next steps (in priority order)

1. Rewrite abstract from scratch as a single paragraph: problem line, MinExplore pipeline, local observation + RL validation, and 1-2 key metrics.
2. Restructure introduction into: context, one-line problem statement, gap, our approach, bullet-point contributions; move dense citations to Related Work.
3. Define and cleanly present the environment pipeline with explicit stages and a pipeline figure that shows real → 2D → 3D → MuJoCo → local observations.
4. Refocus figures on pipeline, environment views, and robot perspective; remove or minimize generic RL training tables.

5. Write a short RL validation section: PPO baseline, reward structure, success/collision outcomes; clearly state RL is only for validation.
6. Add a limitations & scope subsection that honestly talks about elevation, physics realism, and when to use MinExplore vs Gazebo.
7. Fix terminology and formatting: define MJCF, remove italic headings, fix two-column layout, and run language cleanup at the very end.

If you paste or upload the current LaTeX/PDF later, I can go line by line with suggested rewrites, figure captions, and even edited section text tailored to your exact draft.



1. <https://groups.google.com/g/ai-robotics/c/HwaU4ZKKulc>
2. <https://2026.ieee-icra.org/workshops-and-tutorials/>
3. <https://pp-blog.paperpal.com/news-updates/product-updates/ieee-citation-style-guide-with-templates>
4. <https://www.scribbr.com/ieee/ieee-paper-format/>
5. [https://owl.purdue.edu/owl/research\\_and\\_citation/ieee\\_style/ieee\\_general\\_format.html](https://owl.purdue.edu/owl/research_and_citation/ieee_style/ieee_general_format.html)
6. <https://www.underleaf.ai/templates/icra>
7. <https://2025.ieee-icra.org/contribute/final-paper-submission-instructions/>
8. <https://ieee-itss.org/2026/04/call-for-participation-workshop-icra-2026-robot-meets-gnss-and-ranging-for-seamless-autonomy/>
9. <https://2025.ieee-icra.org/event/safely-leveraging-vision-language-foundation-models-in-robotics-challenges-and-opportunities/>
10. <https://saferobotics.princeton.edu/ptas-icra25/awards>
11. <https://sites.google.com/view/icra-2025-ws-sassr/>
12. <https://2026.ieee-icra.org/contribute/call-for-late-breaking-results/>
13. <https://2025.ieee-icra.org/workshops-and-tutorials/>
14. <https://2026.ieee-icra.org/contribute/>
15. <https://iscicra25.github.io>